

ATUL SHARMA

Data Scientist | AI/ML Engineer | Generative AI Specialist

+91-9528610870 | atulsharmagbn1906@gmail.com | linkedin.com/in/atul-sharma-0ab286304 |
atulsharma1906.github.io/Portfolio

PROFESSIONAL SUMMARY

Results-driven Data Scientist with 4 years of experience in Machine Learning, Deep Learning, and Generative AI. Proven track record building enterprise-grade RAG applications, multi-agent AI workflows, and scalable AI systems using LangChain, LangGraph, GPT-4, Mistral, and FastAPI. Experienced in developing AI copilots, chatbot systems, and predictive models. Proficient in FAISS, Pinecone, Docker, Kubernetes, and AWS cloud-native architectures.

TECHNICAL SKILLS

Programming Languages: Python, SQL, MySQL, PostgreSQL

Machine Learning: Scikit-learn, XGBoost, Logistic Regression, Random Forest, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning, SMOTE, EDA, Predictive Analytics

Deep Learning & NLP: TensorFlow, Keras, PyTorch, CNN, RNN, LSTM, Text Classification

Generative AI & LLMs: GPT-4, Mistral 7B, LLaMA 3, BERT, Prompt Engineering, RAG, Agentic AI, Multi-Agent Systems, Conversational AI, Tool Calling, Function Calling

Agentic AI & Orchestration: LangGraph, LangChain Agents, Workflow Orchestration, Dynamic Tool Routing, Stateful AI Agents, Conversational Memory, Autonomous AI Workflows

RAG & Retrieval Systems: Semantic Search, Hybrid Retrieval, BM-25, Cross-Encoder Reranking, Embedding Models, Recursive Chunking, Metadata Filtering, Vector Search

GenAI Frameworks & APIs: LangChain, LangGraph, Hugging Face Transformers, OpenAI API, Hugging Face Hub

Vector Databases & Caching: Pinecone, FAISS, Redis

Backend & APIs: FastAPI, Flask, REST APIs, Async Processing

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, Cloud-Native Deployment

Monitoring & Observability: LangSmith, Prometheus, Grafana

Data Processing: Pandas, NumPy, OCR, Data Ingestion Pipelines

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit

PROFESSIONAL EXPERIENCE

Data Scientist / Generative AI & Machine Learning Engineer | Orbex Technologies Pvt. Ltd, Delhi *June 2022 – Present*

- Developed enterprise AI Copilot platform using LangGraph and RAG architecture, enabling intelligent interaction with structured and unstructured enterprise data.
- Built hybrid retrieval systems combining vector search, BM25 retrieval, and cross-encoder reranking to improve retrieval precision and reduce hallucinations.
- Developed NL-to-SQL workflows for conversational enterprise analytics, enabling non-technical users to query databases using natural language.
- Implemented Redis-based conversational memory and FastAPI backend services for scalable, stateful AI applications.
- Deployed AI infrastructure on Docker, Kubernetes, and AWS for production-grade scalability and reliability.
- Monitored retrieval quality, latency, and agent performance using LangSmith, Prometheus, and Grafana observability pipelines.
- Collaborated cross-functionally on AI integration, API development, deployment optimization, and system monitoring.

PROJECTS

Enterprise Multi-Agent AI Copilot Platform | GenAI – Agentic AI Enterprise Solution

Developed an enterprise-grade AI Copilot platform enabling intelligent interaction with structured (SQL databases) and unstructured enterprise data (PDF, CSV, TXT, multimodal documents) using natural language. The platform combines RAG, multi-agent orchestration, hybrid retrieval, and cloud-native deployment to deliver accurate, context-aware, and scalable enterprise insights.

- Designed and implemented a multi-agent AI architecture using LangGraph for dynamic workflow orchestration and intelligent tool routing.
- Built an enterprise RAG pipeline including document ingestion, recursive chunking (500 chunk size, 10% overlap), embedding generation, semantic retrieval, metadata filtering, and contextual grounding.
- Implemented hybrid retrieval architecture combining vector search, BM25 keyword retrieval, and cross-encoder reranking to improve retrieval precision and reduce hallucinations.
- Integrated advanced LLMs and embedding models for enterprise-grade reasoning, semantic understanding, and contextual response generation.
- Developed Natural Language to SQL workflows enabling non-technical users to query enterprise databases using conversational language.
- Implemented Redis-based conversational memory for maintaining contextual and stateful multi-turn interactions.
- Built FastAPI-based backend services for AI orchestration, agent workflows, authentication, retrieval APIs, and asynchronous processing.
- Worked on multimodal AI workflows including OCR and image-based enterprise document understanding.
- Implemented evaluation and observability pipelines using LangSmith, Prometheus, and Grafana for monitoring hallucination rate, routing accuracy, retrieval quality, latency, and agent performance.
- Collaborated with DevOps teams on Dockerization, Kubernetes orchestration, AWS deployment workflows, and scalable cloud-native infrastructure.
- Optimized enterprise AI workflows using async task execution, caching strategies, and incremental indexing mechanisms.

Key Features: Multi-agent AI orchestration, RAG, Hybrid retrieval + reranking, NL-to-SQL conversion, Multimodal AI workflows, Redis conversational memory, Async task execution, Cloud-native deployment, Agentic AI dynamic tool selection.

Business Impact: Improved enterprise data accessibility; enabled self-service analytics for non-technical users; reduced retrieval latency through hybrid retrieval optimization; automated enterprise knowledge-access workflows.

Technologies: Python, FastAPI, LangGraph, LangChain, GPT-4o, OpenAI Embeddings, Pinecone, Redis, Streamlit, Celery, Docker, Kubernetes, AWS, BM25, OCR, SQL, Pandas, Transformers, Prometheus, Grafana

Context-Aware AI Chatbot | RAG + GenAI System

Intelligent conversational AI chatbot built using Generative AI and RAG architecture, delivering accurate, context-grounded responses through optimized retrieval pipelines and multi-turn memory management.

- Built a production RAG chatbot using GPT-4 and LangChain, integrating Pinecone vector search with top-k retrieval and metadata filtering to serve accurate, context-grounded responses.
- Designed optimized chunking strategies and embedding pipelines, reducing average retrieval latency by ~25% versus naive chunking approaches.
- Implemented Redis session memory for multi-turn conversations and deployed RESTful APIs via FastAPI with rate limiting and error handling.
- Optimized retrieval performance and response latency through systematic prompt tuning and retrieval parameter refinement.

Technologies: Python, FastAPI, GPT-4, LangChain, Pinecone, Redis, Docker, AWS

Customer Churn Prediction System | End-to-End Machine Learning System

Developed an end-to-end machine learning system to predict customer churn based on behavioral and transactional data, enabling proactive retention strategies and reducing revenue loss.

- Collected and processed large-scale customer data including usage patterns, transactions, and engagement metrics.
- Performed data cleaning, EDA, and feature engineering (tenure, recency, frequency, inactivity trends).
- Built and compared Logistic Regression, Random Forest, and XGBoost models; achieved 0.90 ROC-AUC with XGBoost through hyperparameter tuning.
- Handled class imbalance using SMOTE and class weighting; identified key churn drivers using feature importance and SHAP analysis.
- Designed automated ML pipeline for preprocessing, training, and evaluation.
- Developed FastAPI-based API for real-time churn prediction with optimized latency and integrated model outputs with a dashboard for business insights.

Business Impact: Enabled early identification of high-risk customers, improving retention strategy efficiency by 25%; helped reduce churn and improve customer lifetime value (CLV).

Technologies: Python, Pandas, NumPy, Scikit-learn, XGBoost, SHAP, FastAPI, Docker, Jupyter Notebook, Git

EDUCATION

Bachelor of Computer Applications (BCA)

2020 – 2023

Kalinga University